Hunter Ellis

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Engineer focused on building intelligent, capable platforms. I am experienced in developing software for autonomous systems, particularly in the fields of estimation, motion planning, and control systems.

Experience

Avionics Research Engineer — **Aug 2025 – Present**
Georgia Tech Research Institute • Applied Systems Lab — *Atlanta, GA*

- Developing C++ software for collaborative autonomous UAV swarms in a component-based software engineering environment.
- Integrating, developing, and benchmarking sampling-based motion planning algorithms over constraint manifolds leveraging The Open Motion Planning Library (OMPL).
- Developed a containerized motion planning service with hybrid planning using graph, sampling, and optimization methods.
- Integrated multi-hypothesis tracking (MHT) software, including prediction and data association algorithms for multi-object tracking (MOT) in decentralized swarms with distributed sensor data.
- Wrote data association algorithms for multi-object tracking (MOT) and vehicle-to-vehicle track consensus.
- Created a world model service for a collaborative-decentralized swarm.
- Software systems integration

Robotics Research / M.S. Computer Engineering — **Aug 2023 – May 2025**
Virginia Tech • Control, Optimization, and Online Learning for Autonomy Lab — *Blacksburg, VA*

- Built a 6-DOF robotic arm and an accompanying ROS2–Gazebo simulation environment for training, evaluating, and deploying custom control algorithms.
- Integrated object detection (YOLOv8), natural language processing, symbolic reasoning, and a DDPG-based reinforcement learning policy for robotic manipulation tasks.
- Aided professors in teaching fundamental concepts in linear systems theory and digital signal processing, including Laplace Transforms, Z-Transforms, system stability, and FIR & IIR filter design.

Thrust Vector Control Intern — **May 2024 – Aug 2024**
Jacobs Space Exploration Group • Thrust Vector Control Test Lab — *Huntsville, AL*

- Developed thrust vector control testing hardware and software for NASA's Active Inertial Load Simulator at the Marshall Space Flight Center.
- Created and ran tests to develop a mathematical model of an electro-mechanical actuator – used Python, MATLAB, and LabVIEW.

Control Research Intern — **Jun 2023 – Aug 2023**
G2ELab • Power Electronics Lab — *Grenoble, France*

- Implemented inverter control methods for 4-leg (microgrid) topologies and tested them in Simulink and HIL.

Naval Research Enterprise Internship Program — **Jun 2022 – Aug 2022**
Naval Surface Warfare Center • Carderock Division — *Bethesda, MD*

- Developed concept hospital sea-train designs, estimated fuel consumption, and electrical power loads.

Skills

Languages: C++, Python, MATLAB/Simulink, C, Lua, LaTeX

Tools & Libraries: ROS2, OMPL, DDS, Qt, Simulink, Docker, Git, FreeRTOS, SolidWorks, Autodesk Inventor (Certified)

Education

M.S. in Aerospace Engineering – *In Progress* — **Dec 2029**
Georgia Tech • Daniel Guggenheim School of Aerospace Engineering — *Atlanta, GA*

- Focus: Flight Mechanics and Control

M.S. in Computer Engineering — **May 2025**
Virginia Tech • Bradley Department of Electrical & Computer Engineering — *Blacksburg, VA*

- Focus: Software and Machine Intelligence
- Research Group: Control, Optimization, and Online Learning for Autonomy Lab at UT Austin

B.S. in Electrical Engineering and Computer Engineering (Double Major) — **May 2024**
Virginia Tech • Bradley Department of Electrical & Computer Engineering — *Blacksburg, VA*

- Focus: Controls, Robotics, and Autonomy
- Design Teams: Solar Car and Human Powered Submarine